

Verifier-Guided Multi-Teacher Distillation for Streaming Tissue Tracking

STIR Challenge 2026 — Team NCT_TSO

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1 Overview

Our model uses a single LiteTracker [1] student, fine-tuned by distilling predictions from an *ensemble of six point trackers*. We do not introduce a new architecture. LiteTracker is the causal, streaming formulation of CoTracker3-Online [2], so it can load a fine-tuned CoTracker3-Online checkpoint directly while retaining constant-memory ring-buffer inference.

The challenge dataset provides an unusual supervision signal: the IR tattoo is annotated at the *start and end* of every clip, but not in the intervening frames. We combine the final annotation with two self-supervised trust signals to *verify* candidate trajectories from several teachers. We then fuse the candidates into a cleaned pseudo-label trajectory and assign a confidence weight to each frame. The student is trained on these weighted trajectories. The same checkpoint serves all three submitted tracks, with a different inference entry point for each track. Or codabse is available at https://github.com/danushkv/STIR2026_challenge

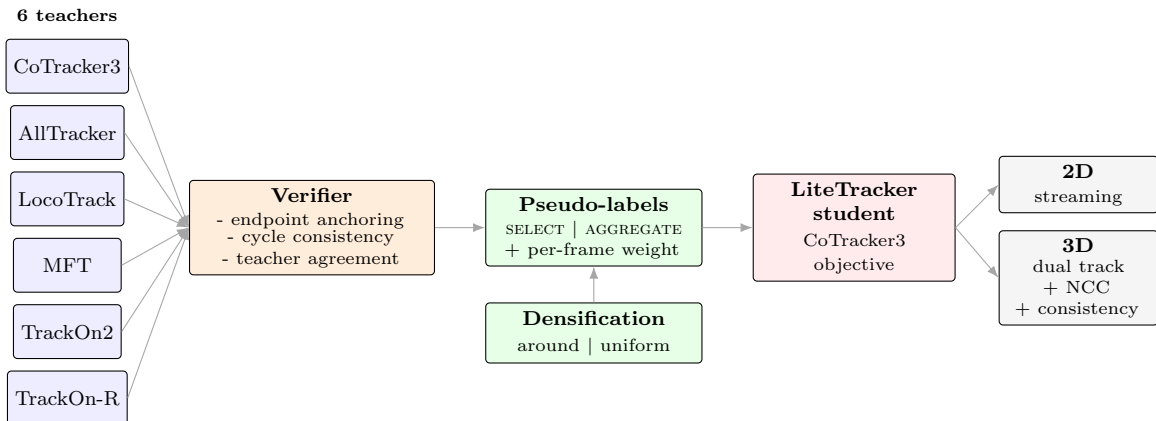


Figure 1: Pipeline. Six teachers produce candidate trajectories. The verifier scores them with one supervised and two self-supervised signals. The resulting pseudo-labels, optionally densified, provide weighted supervision for one LiteTracker student serving both tracks.

2 Method

2.1 Teacher ensemble

Figure 1 summarises the pipeline. We ran six trackers over the STIROrig dataset [8], storing raw trajectories per clip: CoTracker3 [2], AllTracker [3], LocoTrack [4], MFT [5], Track-On2 [6], and Track-On-R [7]. We selected trackers with different architectures, including correlation-based transformers, flow-chaining methods, and recurrent models. Their different failure modes make agreement across teachers a useful confidence signal.

2.2 Verifier-guided pseudo-labels

Raw teacher tracks are noisy. We score every teacher at every frame using three trust signals:

1. **Endpoint anchoring** (STIR-specific). The true location at the final frame is known from the IR tattoo. A teacher whose trajectory misses that endpoint is untrustworthy for the whole clip, giving a per-teacher *global*

trust multiplier.

2. **Cycle consistency** (self-supervised). Track forward to frame T , then backward from that endpoint toward frame 0. Disagreement between the two passes marks unreliable frames, giving a per-teacher *per-frame* reliability.
3. **Teacher agreement** (self-supervised). At each frame, teachers close to the robust median are more trustworthy than outliers.

These combine into a per-frame score per teacher. We compared two fusion modes: SELECT, which picks the single best teacher per frame, and AGGREGATE, a confidence-weighted blend of all teachers. The final model uses AGGREGATE mode.

2.3 Dense pseudo labels

To increase supervision density, we add query points to frame 0 and track them with a single teacher. Rejection sampling enforces a minimum separation from existing and newly sampled points. Since these points have no annotated endpoint, they receive lower weights. We compare uniform sampling (*extra*) with sampling near the annotated tattoos (*around*). Refer to Fig. 2 for an example.

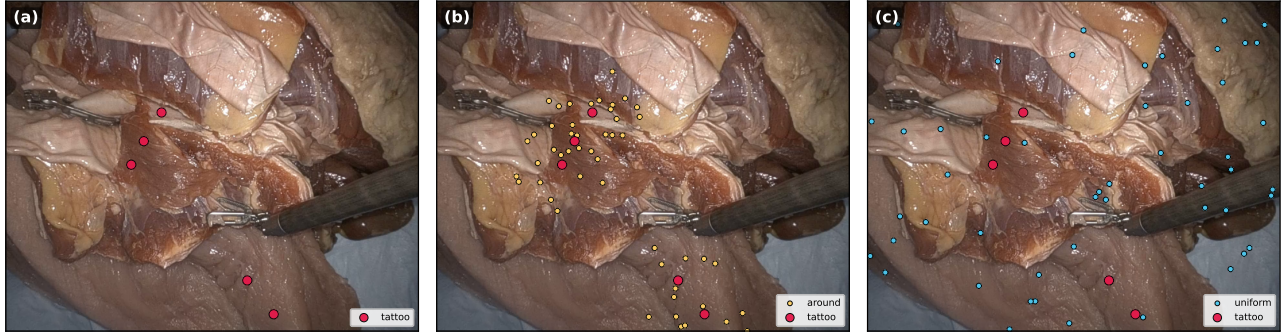


Figure 2: Query points on frame 0 of a STIR clip: (a) IR tattoo centres, the only annotated supervision available. (b) Plus points sampled in a disk around each tattoo. (c) Plus points sampled uniformly across the frame.

2.4 Student training

We initialise the student from LiteTracker and retain CoTracker3’s training objective:

$$\mathcal{L} = 0.05 \mathcal{L}_{\text{coord}} + \mathcal{L}_{\text{vis}} + \mathcal{L}_{\text{conf}},$$

where the three terms supervise coordinates, visibility, and confidence. The coordinate loss is an iteration-weighted ℓ_1 loss with $\gamma = 0.8$, applied only to visible frames. Verifier confidence scales the validity mask, reducing the contribution of unreliable pseudo-labels without discarding them.

We train the transformer module while keeping the initialised feature CNN fixed. Training uses a learning rate of 2×10^{-5} for 45 epochs, 16-frame windows, four refinement iterations, a resolution of 384×512 , and up to 64 frames and 384 points per clip. Pseudo-labels are collected with temporal stride `skip=5`. We also evaluate a threshold-aware loss based on $\sigma((\theta - \text{dist})/\tau)$ over the challenge thresholds.

2.5 Stereo (3D) inference

The 3D track runs *two* instances of the same student in parallel on the left and right streams and then triangulates them. The 2026 harness supplies both images every frame but only *left* query coordinates, so the right-view correspondents must be recovered from pixels: we do this once, on frame 0, by normalised cross-correlation along the epipolar row (the pair is rectified to a common principal point, so the search is 1-D), with a parabolic sub-pixel refinement.

Because disparity is a *difference* of two independently tracked points, their errors add and a single bad triangulation is otherwise emitted as a confident 3D point. We therefore add a **geometric consistency check**: the queries lie on a grid over a smooth tissue surface, so disparity varies smoothly, and a point departing grossly from its neighbours’ median disparity is replaced by that median. This repairs rather than rejects, because ATA scores position regardless of predicted visibility.

Table 1: Ablations evaluated with streaming inference and one refinement iteration. AGGREGATE fuses teachers using confidence-weighted averaging, whereas SELECT uses the highest-scoring teacher at each frame. *Around* samples extra points near annotated tattoos, *random* samples them uniformly over the image, and *--* uses no extra points. A checkmark in THR. enables the threshold-aware loss. All models start from LiteTracker unless marked *, which denotes initialization from the 2025 winning checkpoint. The static-point baseline predicts that the query never moves.

Fusion	Extra points	Thr.	Epoch	2D	3D
Aggregate	—	—	44	.8085	.7339
Aggregate	—	—	40	.8077	.7260
Aggregate	around	✓	30	.8060	.7137
Aggregate	around	—	30	.8043	—
Aggregate	around	—	44	.8034	.7172
Select	around	✓	44*	.8017	.7084
Select	random	✓	40*	.8017	.7233
Select	around	—	42*	.8034	.7172
Select	random	✓	42	.8009	.7313
Select	random	✓	30	.7991	.7207
Select	random	—	40	.7974	.7278
Select	—	—	45	.7974	.7269
Static point				—	.6115

Table 1 lists every configuration we trained. We noticed the aggregate fusion method to score better than the select strategy. Similar behaviour was noticed for 3D tracking, with the results being clearly seperable over the methods. Neither threshold-aware loss nor around-point densification did help. Both these configurations landed inside the noise band. We report this as a negative result rather than selecting the variant that happened to score highest.

We selected **agg**, **epoch 44** because it is the only configuration leading *both* columns, has the best 2D median error (4.55 px), and never diverged on any clip.

3 Limitations

Per-frame AJ/ATA/OA could not be measured as the STIROrig dataset annotates only the first and last frame, so all our numbers are endpoint proxies, and no released split carries the 2026 `annotations.json`. Visibility is therefore unvalidated, and the stereo visibility gate is deliberately conservative, which may cost occlusion accuracy.

References

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